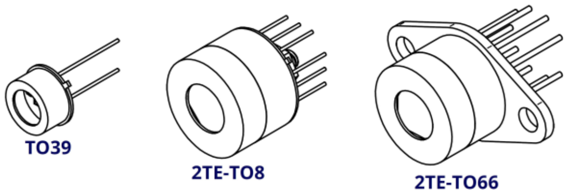
PVM-10.6 DETECTOR SERIES

DATASHEET

HgCdTe room temperature and thermoelectrically cooled photovoltaic multi-junction infrared detectors



FEATURES

- Spectral range: 2.0 to 13.0 µm
- Large active area
- Back-side illuminated
- Fast response
- No minimum order quantity required
- Detector PVM-10.6-1x1-TO39-NW-90 is a Selected product

RELATED PRODUCTS

- LabM-I-10.6 detection module
- UM-I-10.6 detection module
- microM-10.6 detection module
- PVIA-10.6-1x1-TO39-NW-36 RoHS-compliant detector
- PVIA-4TE-10.6-1x1-TO8-wZnSeAR-36 RoHS-compliant detector

APPLICATIONS

- Gas detection, monitoring and analysis: SO_2, NH_3, SF_6
- CBRN threats detection
- CO_2 laser measurements: power monitoring and control, beam profiling and positioning, calibration
- Free-space optical communication
- FTIR spectroscopy
- Bacteria identification in medicine
- Dentistry
- Glucose sensing

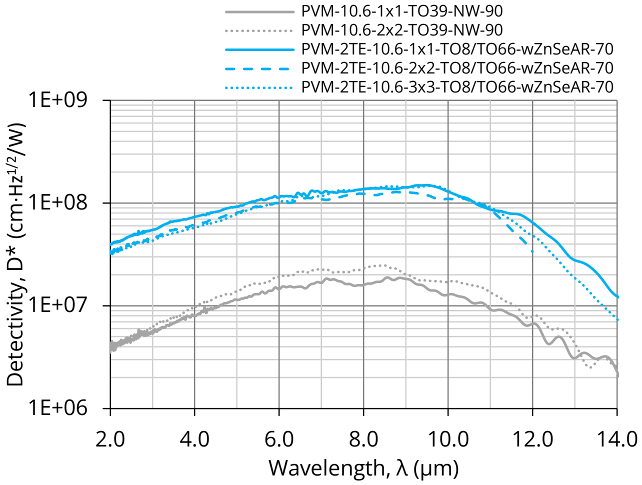
SERIES DESCRIPTION

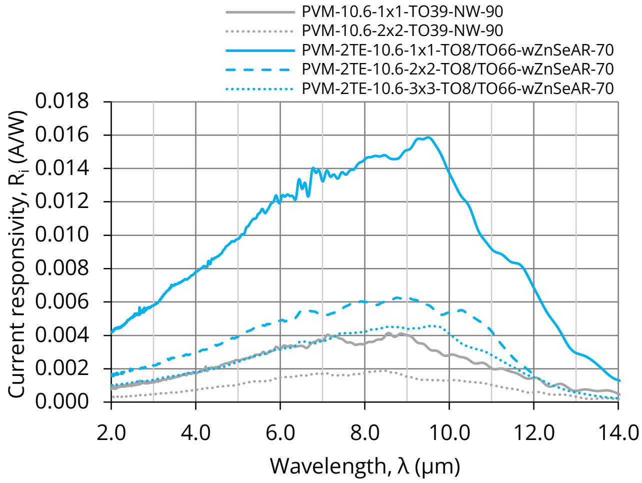
Detector symbol	Cooling	Temperature sensor	Active area, A, mm×mm	Optical immersion	Package	Acceptance angle, Φ, deg.	Window
PVM-10.6-1x1-TO39-NW-90	no	n/a	1×1	no	TO39 (3 pin)	~90	no
PVM-10.6-2x2-TO39-NW-90			2×2				
PVM-2TE-10.6-1x1-TO8-wZnSeAR-70	2TE T_{chip}≅230K	thermistor	1×1		2TE-TO8	~70	wZnSeAR (3 deg. zinc selenide, anti-reflection coating)
PVM-2TE-10.6-1x1-TO66-wZnSeAR-70					2TE-TO66		
PVM-2TE-10.6-2x2-TO8-wZnSeAR-70			2×2		2TE-TO8		
PVM-2TE-10.6-2x2-TO66-wZnSeAR-70					2TE-TO66		
PVM-2TE-10.6-3x3-TO8-wZnSeAR-70			3×3		2TE-TO8		
PVM-2TE-10.6-3x3-TO66-wZnSeAR-70					2TE-TO66		

SPECIFICATION (T_{amb} = 293 K, V_b = 0 V)

Detector symbol	Cut-on wavelength (10%)	Peak wavelength	Specific wavelength	Cut-off wavelength (10%)	Detectivity		Current responsivity			Time constant	Dynamic resistance	
	λ_{cut-on}	λ_{peak}	λ_{spec}	$\lambda_{cut-off}$	D*(λ_{peak}, 20kHz)	D*(λ_{spec}, 20kHz)	$R_i(\lambda_{peak})$	$R_i(\lambda_{spec})$		τ	R_d	
	µm	µm	µm	µm	cm·Hz$^{1/2}$/W	cm·Hz$^{1/2}$/W	A/W	A/W		ns	Ω	
	Typ.	Typ.	Typ.	Typ.	Typ.	Min.	Typ.	Min.	Typ.	Typ.	Min.	Typ.
PVM-10.6-1x1-TO39-NW-90	2.0	8.5±1.0	10.6	12.0	2.0×10^7	1.0×10^7	0.004	0.002	0.0025	1.5	30	50
PVM-10.6-2x2-TO39-NW-90							0.002	0.001	0.0015			
PVM-2TE-10.6-1x1-TO8-wZnSeAR-70		9.0±1.0		13.0	1.5×10^8	1.0×10^8	0.015	0.01	0.012	4	70	120
PVM-2TE-10.6-1x1-TO66-wZnSeAR-70												
PVM-2TE-10.6-2x2-TO8-wZnSeAR-70							0.007	0.005	0.006			
PVM-2TE-10.6-2x2-TO66-wZnSeAR-70												
PVM-2TE-10.6-3x3-TO8-wZnSeAR-70							0.0045	0.003	0.004			
PVM-2TE-10.6-3x3-TO66-wZnSeAR-70												

SPECTRAL RESPONSE (Typ., T_{amb} = 293 K)





MECHANICAL LAYOUT AND PINOUT

- TO39(3p)-NW, PV detector technical drawing
- 2TE-TO8(12p)-wW, PV detector technical drawing
- 2TE-TO66(9p)-wW, PV detector technical drawing

RECOMMENDED AMPLIFIERS

Detector symbol	Amplifier type
PVM-10.6-1x1-TO39-NW-90	SIP-TO39 series
PVM-10.6-2x2-TO39-NW-90	
PVM-2TE-10.6-1x1-TO8-wZnSeAR-70	AIP series PIP series MIP series SIP-TO8 series
PVM-2TE-10.6-2x2-TO8-wZnSeAR-70	
PVM-2TE-10.6-3x3-TO8-wZnSeAR-70	

ABSOLUTE MAXIMUM RATINGS

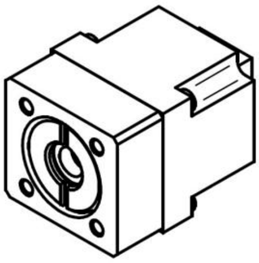
Parameter	Test conditions, remarks	Value	Unit
Ambient operating temperature, T_{amb}	Operation at T_{amb} >30°C may increase the active element temperature and reduce the performance of the detector below specified parameters	-20 to 30	°C
Storage temperature, T_{stg}		-20 to 50	°C
Soldering temperature	Within 5 s or less	≤300	°C
Storage humidity	No dew condensation	10 to 90	%
Maximum incident optical power density	Continuous wave (CW) or single pulses >1 µs duration	100	W/cm^2
	Single pulses <1 µs duration	1	MW/cm^2
Maximum bias voltage, $V_{b\ max}$	No bias voltage needed	-	-
Maximum TEC voltage, $V_{TEC\ max}$	2TE	1.0	V
Maximum TEC current, $I_{TEC\ max}$	2TE	1.2	A

Stresses beyond those listed under absolute maximum ratings may cause permanent damage to the device. Constant or repeated exposure to absolute maximum rating conditions may affect the quality and reliability of the device.

SIP-TO39 SERIES

DATASHEET

Small-size, for uncooled detectors, transimpedance amplifiers



FEATURES

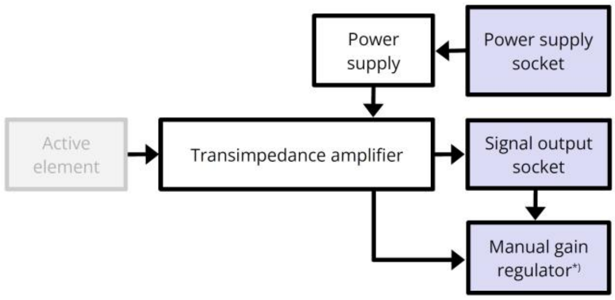
- Compatible with VIGO uncooled IR detectors in the TO39 package
- Frequency bandwidth: up to 250 MHz
- Small size
- AC or DC coupled
- External power supply required
- Compatible with optical accessories
- Adjustable gain (optional, modules with a frequency bandwidth of up to 100 MHz)

CODE DESCRIPTION

Type		f_{lo}, Hz		f_{hi}, Hz		Detector package		Gain adjustment
SIP	–	DC 10 100 1k 10k	–	100k 1M 10M 100M 250M	–	TO39	–	G*) (with gain adjustment) NG (without gain adjustment)

*) Only for SIP amplifiers with $f_{hi} \leq 100$ MHz

SCHEMATIC DIAGRAM



*) Only for the SIP-xx-xx-TO39-G version.

INCLUDED ACCESSORIES

- 1 pc of MMCX-BNC cable
- 1 pc of AMP2x4-DB9 cable

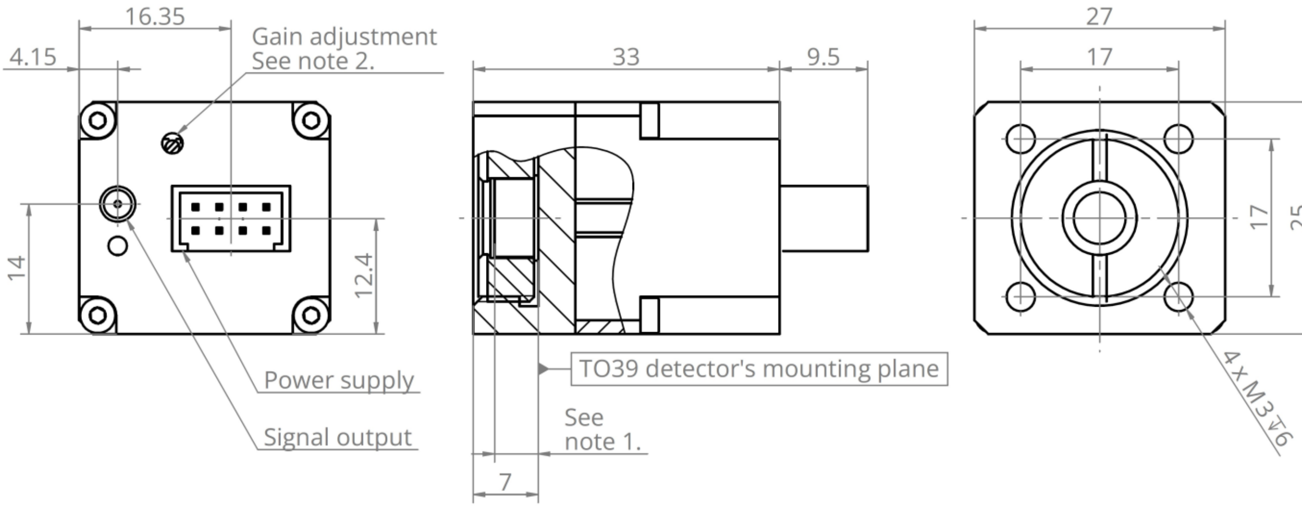
DEDICATED ACCESSORIES

- PPS-03 power supply series

SPECIFICATION (T_{amb} = 293 K)

Parameter	Conditions, remarks	Value	Unit
Low cut-off frequency, f_{lo}		DC, 10, 100, 1k, 10k	Hz
High cut-off frequency, f_{hi}		100k, 1M, 10M, 100M, 250M	Hz
Transimpedance, K_i	Tunable, only the SIP-xx-xx-TO39-G version	up to 100	kV/A
Transimpedance range, $K_{i\,max}/K_{i\,min}$	Depending on the f_{hi}, only the SIP-xx-xx-TO39-G version	up to 2	-
Output impedance, R_{out}		50	Ω
Output voltage swing, V_{out}	$f_{hi} \leq 1$ MHz, R_{load} = 1 MΩ	±10	V
	$f_{hi} > 1$ MHz, R_{load} = 50 Ω	±1	
Output voltage offset, V_{off}		max. ±20	mV
Power supply voltage, V_{sup}	$f_{hi} \leq 1$ MHz, R_{load} = 1 MΩ	±15	V
	$f_{hi} > 1$ MHz, R_{load} = 50 Ω	±9	
Power supply current, I_{sup}		max. ±50	mA
Weight		52	g

MECHANICAL LAYOUT (Unit: mm)

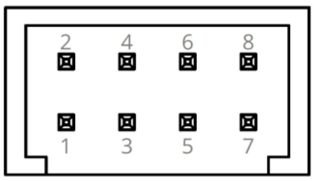


Notes:

1. TO39 detector dimensions in the TO39 package technical drawings.
2. Only for the SIP-xx-xx-TO39-G version.

POWER SUPPLY SOCKET PINOUT

AMPMODU 280389-2 (male)



Pin No.	Symbol	Function
1	-Vsup	Power supply input (-)
2	NC	Not connected
3	GND	Ground
4	NC	Not connected
5	GND	Ground
6	NC	Not connected
7	+Vsup	Power supply input (+)
8	NC	Not connected

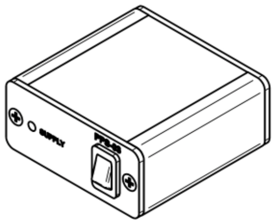
ABSOLUTE MAXIMUM RATINGS

Parameter	Test conditions, remarks	Value	Unit
Ambient operating temperature, T_{amb}		10 to 30	°C
Storage temperature, T_{stg}		-20 to 50	°C
Humidity	No dew condensation	10 to 90	%

Stresses beyond those listed under Absolute maximum ratings may cause permanent damage to the device. Constant or repeated exposure to absolute maximum rating conditions may affect the quality and reliability of the device

PPS-03 SERIES

DATASHEET



Power supplies

VIGO IR DETECTION MODULES THAT CAN OPERATE WITH PPS-03 SERIES

- microM-10.6
- IR detection modules with uncooled detectors in the TO39 package and the SIP-TO39 series amplifiers

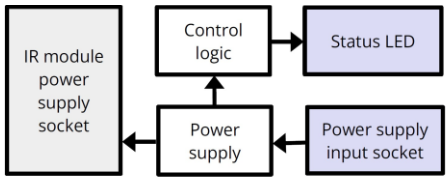
INCLUDED ACCESSORIES

- 1 pc of AC adaptor

CODE DESCRIPTION

Type		Output voltage, V_{DC}
PPS-03	–	09
		15

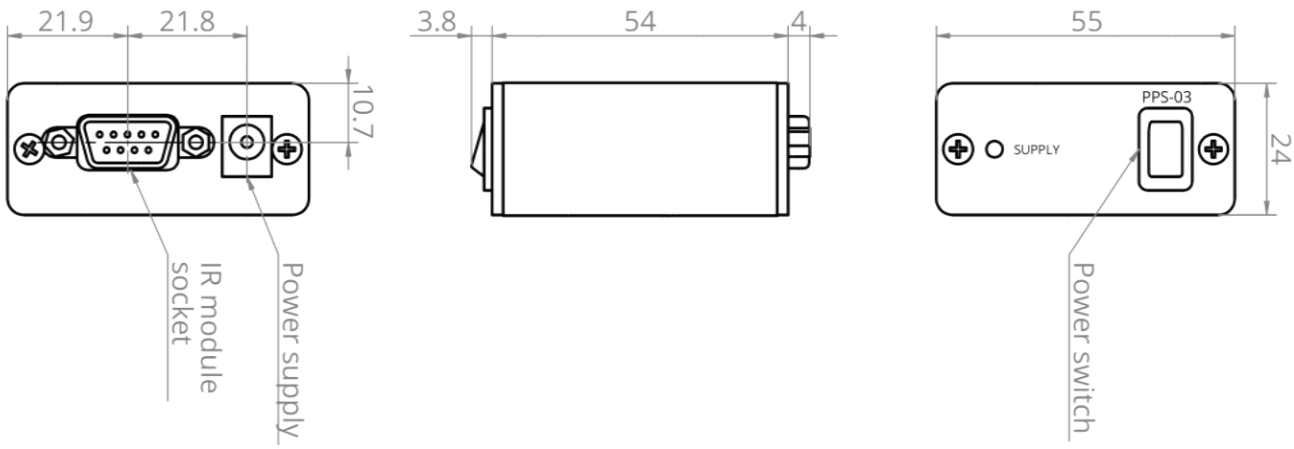
SCHEMATIC DIAGRAM



SPECIFICATION (T_{amb} = 293 K)

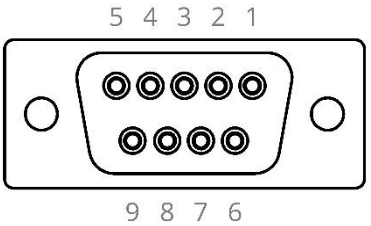
Parameter	Test conditions, remarks	Value			Unit
		Min.	Typ.	Max.	
Input power supply voltage range		9.0	-	16.0	V_{DC}
IR detection module power supply output voltage	PPS-03-09	-	±9	-	V_{DC}
	PPS-03-15	-	±15	-	
IR detection module power supply output current		-	-	±100	mA
Weight		-	90	-	g

MECHANICAL LAYOUT (Unit: mm)



IR DETECTION MODULE SOCKET PINOUT

D-SUB 9 PIN



Pin No.	Symbol	Function
1	NC	Not connected
2	NC	Not connected
3	GND	IR module power supply ground
4	NC	Not connected
5	NC	Not connected
6	-Vsup	IR module power supply output (-)
7	NC	Not connected
8	NC	Not connected
9	+Vsup	IR module power supply output (+)

ABSOLUTE MAXIMUM RATINGS

Parameter	Test conditions/remarks	Value	Unit
Ambient operating temperature, T_{amb}		5 to 45	°C
Storage temperature, T_{stg}		-20 to 70	°C
Humidity	No dew condensation	10 to 90	%

Stresses beyond those listed under Absolute maximum ratings may cause permanent damage to the device. Constant or repeated exposure to absolute maximum rating conditions may affect the quality and reliability of the device